

Semester 3 Mathematics - Graded Assignment 1

Submission Deadline: October 15, 2026 | Max Marks: 50 | Student: Pranjal Singh

Question 1: Eigenvalues and Spectral Decomposition (15 Marks)

Given matrix $M = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$, compute the characteristic polynomial $\det(M - \lambda I)$.

Verify that $\lambda = 8$ is an eigenvalue and find its corresponding unit eigenvector.

Find the remaining eigenvalues and verify that eigenvectors of distinct eigenvalues are orthogonal.

Question 2: Discrete Probability & Markov Chains (15 Marks)

A server transitions between states: Healthy (H), Warning (W), and Critical (C).

Transition matrix $P = \begin{bmatrix} 0.8 & 0.15 & 0.05 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$.

Compute the stationary distribution π such that $\pi * P = \pi$, with $\sum(\pi) = 1$.

Question 3: Graph Connectivity and Spanning Trees (20 Marks)

Prove that any connected undirected graph with n vertices and $n - 1$ edges is a tree.

Apply Kruskal algorithm to find the Minimum Spanning Tree (MST) on the given network topology.